Calculate gravitational force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate a gravitational interaction with consistent units.

SUCCESS CRITERIA - I can...

1. Identify both masses and the center separation.
2. Square the distance before dividing.
3. Check units and the scale of the result.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “separation” and “mass” to explain your answer. Sentence starters are optional.

separation: distance between the specified objects or centers

mass: a measure of how strongly an object resists acceleration

inverse square: proportional to one divided by distance squared

01 Warm-up: bring back an earlier idea

A ball begins with downward velocity **3 m/s** and falls **2 s**.
1. Use downward positive, g=**10 m/s²** and no drag.
2. **Find final velocity** and displacement.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

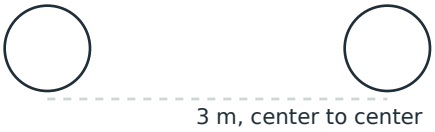
02 Worked example: explain the prediction

Reference: $F = Gm_1m_2/r^2$; $G = \mathbf{6.67\times10^{-11}\ N\cdot m^2/kg^2}$.
r is center-to-center separation. CPA may use the supplied numerator.

Worked reasoning: In the supplied gravity calculation, Gm_1m_2=**18 N·m²** and r=**3 m**. Square first: r^2=**9 m²**, so F=**18/9**=**2 N**, attractive.

Your turn: 1. **Sketch the two centers**, label r and **add the two force arrows**.
2. **Show how the units reduce** to newtons.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



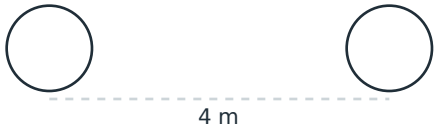
Mass 1 Mass 2

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

For spheres **4 m** apart center to center, a student substitutes r=**2 m** because each is **2 m** from the midpoint. **Name the error**.

Sentence starter: The step I would change is ... . I would change it because I found ... .



Sphere A Sphere B

Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

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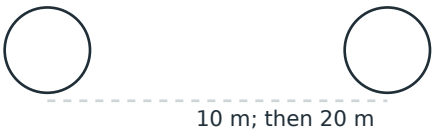
Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

Two idealized spherical space objects have masses **2.0 × 10⁵ kg** and **3.0 × 10⁵ kg**. Their centers are **10 m** apart.
1. Use G = **6.67 × 10⁻¹¹ N·m²/kg²**.
2. **Label the diagram** and **calculate their attraction**.
3. Repeat at **20 m**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

Why can a tiny force between classroom-sized objects be physically real even if you cannot feel it?

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** box slides left at **6 m/s** on a rough horizontal floor after release. **Draw forces** and **explain what follows about vertical** and horizontal acceleration.
Draw the forces on the chosen object and **label their sources**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

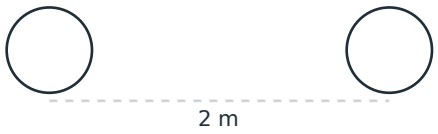
Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

Two **1000 kg** spheres have centers **2 m** apart.
1. Use G = **6.67 × 10⁻¹¹ N·m²/kg²**.
2. **Calculate the attraction**.
3. A classmate divides by r instead of r²: by what factor does that change the numerical answer?

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A pair has force **0.12 N** at **2 m**. Without finding either mass, calculate Gm_1m_2 and the separation for a force of **0.03 N**.

Sentence starter: I found … [unit]. This value means … in this situation.

Separation (m)	Force (N)
___	___
___	___

Marzano check - circle: 1 2 3